

Sahil Undale

Mumbai, India - 400006 — sahil22undale@gmail.com — 7559292204
linkedin.com/in/sahil-undale — github.com/sudosahil — sahilundale.in

PROFESSIONAL SUMMARY

Software developer and AI tools specialist who builds full-stack products and AI-powered automations end to end. Experienced in evaluating AI models, prompt engineering, and integrating LLMs and OCR into real workflows, alongside shipping production applications with Next.js, TypeScript, Node.js, and PostgreSQL. Looking to help turn AI capabilities into products people use.

EXPERIENCE

Self-Employed Freelancer

Mumbai & Pune, India

Freelance Software Product Developer — AI Automation & Product Implementation

2025–Present

- Delivered end-to-end web applications for 5+ businesses and startups, handling everything from UI/UX design and database architecture to final deployment.
- Built AI automations with n8n and the Google Gemini API that source local-business leads, audit their websites, draft personalized pitches, and send throttled outreach, with deduplication on domain, phone, and place ID.
- Engineered full-stack operational tools, including a custom Razorpay-integrated booking platform for Bombay Gaming Co. and a comprehensive admin dashboard with role-based access for Siddhi Coaching Classes.
- Built conversion-optimized marketing engines for clients like TheoremLabs, routing leads directly to WhatsApp and significantly improving local search visibility.
- Developed high-performance architectures using Next.js, TypeScript, Node.js, and PostgreSQL, consistently hitting 95+ Lighthouse scores for client sites.
- Ensured application reliability and zero-downtime deployments by implementing robust automated testing pipelines using Jest, Testing Library, and MSW.
- Maintain a personal portfolio at sahilundale.in showcasing shipped client products and AI automation work.

Next Gen Community (NGC)

Mumbai, India

Research Analyst Intern

January 2025–April 2025

- Researched and evaluated AI models across multiple use cases through hands-on testing and experimentation.
- Designed and refined effective prompts for different models to assess performance, limitations, and optimal applications.
- Authored structured research reports translating complex AI model behavior into clear, readable documentation and best-fit real-world use cases based on empirical testing.

Atraya Technologies

Remote, India

Software Developer

May 2025–August 2026

- Developing an end-to-end e-Governance and project management application serving as a centralized platform for interaction between government departments, citizens, and contractors.
- Engineering workflow-based modules for project tracking, comprehensive bill management, RA bill clearance, and LOC approvals.
- Contributing to a fully integrated HRMS module featuring automated leave calculation, biometric attendance tracking, and statutory compliance management.
- Building customized, role-based dashboards for staff and contractors to enable transparent tracking of project expenditures and fund releases.
- Developing scalable cloud-based application architectures utilizing Microsoft MVC 6 framework and Angular with WebAPI integrations.

BloomBox - Entrepreneurship Cell of KJSCE

Mumbai, India

Events and Operations Member

June 2025–Present

- Managed on-ground logistics and execution for large-scale entrepreneurship events, successfully coordinating crowd flow and operations for 500 to 800 attendees.
- Structured and executed multi-stage event workflows, including directing a dedicated qualifier round leading up to a large-scale main event.
- Coordinated corporate speaker invitations and streamlined logistics for major initiatives, including the E-Summit, securing participation from industry founders and professionals.
- Managed automated outreach and formal communications to build a strong network of guest speakers while collaborating with cross-functional student teams to hit operational deadlines.

PROJECTS & HACKATHONS

NAKSH - Safety-Aware Pedestrian Routing App — Smart India Hackathon (SIH) Final Round 2026

- Engineered a safety-aware pedestrian routing Android app using Flutter and Python/FastAPI that computes the safest walking routes rather than just the fastest options.
- Implemented a modified safety-weighted Dijkstra algorithm using PostgreSQL, PostGIS, and pgRouting to successfully process 455,251 OpenStreetMap segments.
- Designed a 5-parameter safety scoring engine calculating live risks based on lighting, footpaths, activity, assistance, and connectivity, achieving a p50 route calculation latency of 18.2 ms.
- Built a privacy-preserving crowdsourced evidence reporting loop utilizing Redis and Celery for async processing while enforcing k-anonymity parameters.

Mini Raft - Replicated Key-Value Database — Java 21, Maven, JUnit 5 2026

- Implemented the Raft consensus algorithm from the paper in Java: randomized-timeout leader election, log replication, and majority-based commit across a 3-node replicated key-value store.
- Persisted term, vote, and log to disk so a crashed node restarts as a follower and repairs its log; demonstrated data surviving a leader crash and automatic re-election.
- Verified election, commit safety, and crash recovery with JUnit 5 tests on a simulated in-process RPC layer.

ExamMitra - Automated Answer Key Generation — Google Drive, OCR, AI Automation 2026

- Built an end-to-end automation that detects question papers as soon as they are uploaded to a Google Drive folder, with no manual trigger.
- Extracts the questions from each paper using OCR, then generates a structured answer key from them.
- Uploads the finished answer key to the corresponding answer-key Google Drive folder, turning a manual grading-prep task into a hands-free workflow.

AI Lead Generation & Outreach Pipeline — n8n, Google Gemini, PostgreSQL (Neon), Docker 2026

- Designed an automated daily pipeline that sources local businesses in Mumbai and Pune, audits their websites with Gemini, and generates personalized pitches for outreach.
- Orchestrated the workflow in n8n (self-hosted in Docker) with a Neon PostgreSQL backend and a custom freelancer dashboard, enforcing deduplication on domain, phone, and place ID.
- Built in daily send caps and randomized delays to keep outreach throttled and human-paced.

PMIS - Program Management Information System — Zaheerabad Industrial Area (NICDC) 2025–2026

- Co-developing a Program Management Information System for the Zaheerabad Industrial Area, a government project by NICDC, to manage its day-to-day project work in one platform.
- Working in a two-developer team under Atraya Technologies, owning delivery of core modules from requirements through implementation.
- Used AI-assisted development workflows (prompting, reviewing, and iterating on generated code) to accelerate building the system.

Idempotent Payments Ledger — Java, Spring Boot, PostgreSQL, Flyway, Docker 2026

- Building a layered payments service with a double-entry ledger, where every transfer writes debit and credit entries that sum to zero.
- Designing idempotency keys so retried requests never double-charge, and using ACID transactions with row locking to keep balances exact under concurrency.
- Planning JUnit 5 and Testcontainers tests, a 1,000-request concurrency load test with k6, and a reconciliation job.

EDUCATION

KJ Somaiya College of Engineering Mumbai, India
B.Tech in Electronics and Computer Engineering July 2024–August 2028

St. Arnolds Central School — Class XII (PCM, CS) Pune, India — April 2024
Vibgyor High, Balewadi — Class X (PCM, CS) Pune, India — April 2021

SKILLS

- **AI & Automation:** AI Automations (n8n), LLM Integration (Gemini API), Prompt Engineering, AI Evaluation
- **Languages & Frameworks:** TypeScript, Python, Java, Spring Boot, Flutter, Next.js, React, Node.js, Tailwind CSS, Angular
- **Backend & Databases:** PostgreSQL, PostGIS, pgRouting, Redis, Celery, Prisma ORM, Supabase
- **Tools:** n8n, Docker, Git, Razorpay, Shopify
- **Core Competencies:** Data Analysis, Technical Writing, System Architecture, Event Coordination